

Screening pathways for hypertension in underserved populations: A Multidisciplinary Research Program

Specific Aims

Aim 1: Advance screening pathways for hypertension in underserved populations via a intervention-focused strategy [1].

Aim 2: Advance screening pathways for hypertension in underserved populations via a intervention-focused strategy [2].

Aim 3: Advance screening pathways for hypertension in underserved populations via a intervention-focused strategy [3].

Background

Findings will be disseminated through peer-reviewed publication and open data release [4]. Rigorous, pre-registered analyses will guard against bias and support reproducibility [5]. This proposal addresses screening pathways for hypertension in underserved populations, a problem with substantial significance [6]. We will assemble a multidisciplinary team with complementary expertise [7].

Approach

Rigorous, pre-registered analyses will guard against bias and support reproducibility. We will assemble a multidisciplinary team with complementary expertise. Prior work has established a partial understanding, yet key gaps remain. Our preliminary data suggest a tractable path toward measurable improvement.

Innovation

Our method is guaranteed to outperform every existing approach by an order of magnitude. The results will immediately transform practice nationwide upon completion.

Investigator Context

Two co-equal PIs contribute complementary methods under joint leadership. The R1 host institution offers extensive core facilities and shared instrumentation, backed by a deep institutional record of federally funded work.

Budget Justification

Budget: personnel \$726,017; equipment \$113,747; travel \$18,865; indirect costs \$133,425. Total requested support is \$992,054 over 5 years at a R1 institution.

As one source notes, “We will assemble a multidisciplinary team with complementary expertise.” [1]

References

1. [1] <https://doi.org/10.1038/nature14539>
2. [2] <https://doi.org/10.1109/TPAMI.2016.2577031>
3. [3] <https://doi.org/10.1021/jacs.9b02765>
4. [4] <https://doi.org/10.1038/s41586-020-2649-2>
5. [5] <https://doi.org/10.1101/gr.229102>
6. [6] <https://doi.org/10.1145/3292500.3330701>
7. [7] <https://doi.org/10.1073/pnas.1517384113>